

Workbook: Blood Vessels

Week 3 workbook · BIO 004 Human Anatomy · Dr. Sharilyn Rennie · Solano Community College

A comprehensive workbook on the structure of arteries, veins, and capillaries; the three tunics; capillary types; venous valves; the major systemic arteries and veins; and clinical anatomy of vessel disease.

How to use this workbook: Read each question carefully and write your answer in the space provided. Work without looking up the answer first. Once you have written your answer, check it against the online workbook by clicking *Reveal* on that question. Star or circle anything you missed and come back to it later in the week.

General vessel anatomy · 4 questions

Q01 · DOK 1 · Recall

Name the three tunics of an artery or vein from inside out and describe each.

Q02 · DOK 2 · Compare

Compare arteries and veins by wall thickness, lumen size, and presence of valves.

Q03 · DOK 1 · Recall

What is a capillary, what is it made of, and what does it do?

Q04 • DOK 2 • Compare

Compare the three types of capillaries by structure and location.

Artery types • 3 questions**Q05 • DOK 1 • Recall**

What are elastic (conducting) arteries, and where are they found?

Q06 • DOK 1 • Recall

What are muscular (distributing) arteries, and where are they found?

Q07 • DOK 1 • Recall

What are arterioles, and why are they sometimes called resistance vessels?

Veins • 3 questions**Q08 • DOK 1 • Recall**

Why do most veins have valves, and where in the body are valves especially important?

Q09 • DOK 2 • Apply

Explain how the skeletal muscle pump moves venous blood toward the heart.

Q10 • DOK 1 • Recall

What is a venous sinus, and where are sinuses found?

Major systemic arteries • 4 questions**Q11 • DOK 1 • Recall**

Name the major segments of the aorta in order.

Q12 • DOK 1 • Recall

Name the three branches of the aortic arch in order.

Q13 • DOK 1 • Recall

Name the major unpaired branches of the abdominal aorta in order superior to inferior.

Q14 • DOK 1 • Recall

What does each common iliac artery split into?

Major systemic veins • 3 questions**Q15 • DOK 1 • Recall**

What do the superior and inferior venae cavae drain, and where do they empty?

Q16 • DOK 2 • Apply

What is the hepatic portal system, and what is its purpose?

Q17 • DOK 1 • Recall

What do the internal and external jugular veins drain?

Clinical reasoning • 3 questions

Q18 · DOK 3 · Reason

Atherosclerosis is buildup of fatty and fibrous plaques in the tunica intima of arteries. Using what you know about vessel anatomy, predict the structural consequences in the affected artery.

Q19 · DOK 3 · Reason

Varicose veins are dilated, tortuous superficial veins, usually in the legs. Predict the underlying anatomical problem and explain why the result is a dilated vein.

Q20 · DOK 3 · Reason

An aortic aneurysm is a localized dilation of the aorta. Predict what is happening to the wall and why rupture is dangerous.
